

Department Overview Department Name: Department of Mechanical Engineering Faculty: Faculty of Engineering & Architecture Accreditation: Fully accredited by the Pakistan Engineering Council (PEC). Core Technological Focus: Providing engineering solutions for product and process development, ranging from micro-component design to massive industrial plants, machinery, and vehicles. Key areas include transportation, healthcare infrastructure, construction, thermal sensors/devices, and robotics. Career Sectors: Graduates possess highly diverse skills, securing career opportunities across sectors such as manufacturing, automotive, power generation, construction, biomedical, aerospace, process industries, resource management, and oil & gas.

Vision and Mission Vision: To be a highly reputed Mechanical Engineering faculty, that offers its own unique blend of technical specializations and research opportunities. Mission: To provide a modern academically stimulating environment that produces socially responsible young professionals who can solve mechanical engineering challenges and serve ethically.

1. Bachelor of Science in Mechanical Engineering (BS Mechanical)

Admission Requirements Academic Background: F.Sc (Pre-Engineering) or ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least 60% marks. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 57 • Minimum Academic Standing: CGPA $\geq$ 2.0

Program Educational Objectives (PEOs) Graduates are expected to exhibit the following attributes 3 to 5 years post-graduation:

1. Provide solutions to the problems associated with design, manufacturing, and energy systems using fundamentals of mechanical engineering.
2. Design and investigate mechanical systems by using analytical and modern techniques for innovation, technological development, and sustainability.
3. Acquire the competency to address real-world problems in socio-technological areas through ethical values and professional skills.

Curriculum Plan (Third Board of Studies Schema) Note: Credit hour listings denote (Theory - Lab).

1st Semester • CHE-102 | Applied Chemistry | Prerequisite: None | Credit Hours: 2-0 • PHY-205 (+L) | Applied Physics | Prerequisite: None | Credit Hours: 2-1 • MATHP-118 | Calculus & Analytical Geometry | Prerequisite: None | Credit Hours: 3-0 • HUM-164 | Fundamentals of English Grammar | Prerequisite: None | Credit Hours: 2-0 • MECH-201L | Mechanical Engineering Drawing | Prerequisite: None | Credit Hours: 0-2

2nd Semester • MECH-101L / CS-111 | Workshop Practice Lab / Intro to Computer Programming | Prerequisite: None / None | Credit Hours: 0-2 / 2-1 • EE-106(+L) | Basic Electronics | Prerequisite: None | Credit Hours: 2-1 • MECH-244 | Engineering Materials & Metallurgy | Prerequisite: None | Credit Hours: 3-0 • MECH-121 | Thermodynamics-I | Prerequisite: None | Credit Hours: 3-0 • HUM-101 / MECH-141 | Islamic Studies / Engineering Mechanics (Statics) | Prerequisite: None / None | Credit Hours: 2-0 / 3-0 • MECH-247L / HUM-268 | Computer-Aided Grid Drawing (LAB) / Comm. Skills | Prerequisite: None / None | Credit Hours: 0-1 / 3-0

3rd Semester • MATHA-214 | Differential Equations | Prerequisite: Calculus & Geometry | Credit

Hours: 3-0 • MECH-241 / 205L | Engineering Mechanics (Dynamics) & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1 • MECH-222 / 242 | Fluid Mechanics - I / Solid Mechanics - 1 | Prerequisite: None / None | Credit Hours: 3-0 / 3-0 • HUM-102 / EE-108L | Pakistan Studies / Electrical Engineering | Prerequisite: None / None | Credit Hours: 2-0 / 2-1

4th Semester • MATHA-221 | Linear Algebra & Transforms | Prerequisite: Calculus & Geometry | Credit Hours: 3-0 • MECH-223 / 223L | Fluid Mechanics II & Lab | Prerequisite: Fluid Mechanics-I | Credit Hours: 3-0 / 0-1 • MECH-262 / ECON-105 | Machine Design - I / Engineering Economics | Prerequisite: None / None | Credit Hours: 3-0 / 2-0 • MECH-221 / 221L | Thermodynamics-II & Lab | Prerequisite: Thermodynamics-I | Credit Hours: 3-0 / 0-1 • HUM-276 | Technical RPT WRT & PT Skills | Prerequisite: None | Credit Hours: 2-0

5th Semester • MECH-243 / 243L | Solid Mechanics - II & Lab | Prerequisite: Solid Mechanics - 1 | Credit Hours: 3-0 / 0-1 • MECH-363L | Computer-Aided Engineering Lab | Prerequisite: None | Credit Hours: 0-1 • MECH-364 | Mechanical Instrumentation & Control | Prerequisite: None | Credit Hours: 2-1 • MATHA-216 | Numerical Analysis | Prerequisite: Linear Algebra & Transforms | Credit Hours: 2-1 • MECH-304 / 305 | Machine Design - II / Manufacturing Processes - I | Prerequisite: None / None | Credit Hours: 3-0 / 3-0

6th Semester • MECH-306 / 306L | Manufacturing Processes - II & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1 • STAT-203 | Applied Statistics | Prerequisite: None | Credit Hours: 3-0 • MECH-321 / 321L | Heat & Mass Transfer & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1 • MECH-421 | Internal Combustion Engines | Prerequisite: None | Credit Hours: 2-0 • MECH-362 / 362L | Mechanics of Machines & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1

7th Semester • ENGG-203 | Maintenance Engineering | Prerequisite: None | Credit Hours: 2-0 • MECH-322 / 322L | Refrigeration & Air Conditioning & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1 • ENVIRON-344 | Health, Safety & Environment | Prerequisite: None | Credit Hours: 1-0 • MECH-ME / MECH-TE | Management Elective / Technical Elective - I | Prerequisite: None / None | Credit Hours: 3-0 / 3-0 • MECH-462 | Project - I | Prerequisite: None | Credit Hours: 0-3

8th Semester • MECH-461 / 461L | Mechanical Vibrations & Lab | Prerequisite: None | Credit Hours: 3-0 / 0-1 • HUM-310 | Professional Ethics & Entrepreneurship | Prerequisite: None | Credit Hours: 2-0 • MECH-422 | Power Plant Engineering | Prerequisite: None | Credit Hours: 3-0 • MECH-TE / MECH-463 | Technical Elective - II / Project - II | Prerequisite: None / None | Credit Hours: 3-0 / 0-3

2. Master of Science Program (MS)

Admission Requirements Academic Background: 16 years of education or an equivalent benchmark qualification (such as a 4-year BS or B.E. degree) from an HEC-recognized university with at least 60% marks under the annual grading setup or a minimum CGPA of 2.5 out of 4.0. (Note: The raw webpage text retains a copy-paste placeholder typo erringly naming Civil/Petroleum engineering fields within this admission block). **Testing:** A valid GAT General score with at least 50% marks.

Degree Requirements • Total Credit Hours: 30 • Total Coursework Requirements: 8 advanced graduate courses. • **Minimum Academic Standing:** CGPA $\geq$ 2.5